10.Develop an application to change the font and color of the text and display toast message when the user presses the button.

AIM:

To develop an Android application that allows the user to change the font and color of text displayed on the screen and display a Toast message when a button is pressed.

ALGORITHM:

1. Initialize the Android Application:

  o Set up a new Android project in Android Studio.

  o Create a MainActivity Java file for the main screen of the application.

2. Design the User Interface (UI):

o In the XML layout file, add the following UI components:

 TextView: For displaying the text whose font and color will be changed.

 Button: To trigger the change in font and color, and show a Toast message.

 Spinner or Color Picker: To let the user choose a color (optional, or a simple button for a predefined color change).

 RadioButtons or a Dropdown Menu to select different font styles (optional).

3. Declare Variables:

o Create references for the TextView, Button, Color Picker (if used), and Radio Buttons (if used to change font style).

o Define variables for the font style, color, and the message to display in the Toast.

4. Set Event Listener for the Button:

o Set an OnClickListener for the Button.

o Inside the onClick() method, write the logic to:

 Change the font of the TextView based on the selected font style.

 Change the color of the TextView based on the selected color.

 Display a Toast message indicating the action has been performed.

5. Implement Font Style Changes:

o Change the TextView's font using setTypeface() to apply a selected font style (e.g., bold, italic).

o You can either use predefined styles (like Typeface.BOLD or Typeface.ITALIC) or let the user choose from a list of font styles.

6. Implement Color Change:

o Change the TextView's text color by using the setTextColor() method and applying the selected color (e.g., Color.RED, Color.BLUE, or a color from the Color Picker).

o If using a color picker or dropdown, get the selected color and set it on the TextView.

7. Show Toast Message:

o Display a Toast message to the user when the button is clicked, informing them that the text style or color has changed. You can also show a message like "Font and color changed!" or any other appropriate message.

8. End the Operation:

o After the changes are made, the application will display the updated TextView with the new font and color, and the Toast message will appear to provide Feedback.

Code:

1. XML Layout (`activity_main.xml`):

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
  xmlns:android="http://schemas.android.com/apk/res/android"
  android:orientation="vertical"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:gravity="center"
  android:padding="20dp">

  <TextView
    android:id="@+id/weatherText"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Weather Information"
    android:textSize="24sp"
    android:textColor="#000000"
    android:padding="20dp"/>
```

```
    <Button
        android:id="@+id/changeBtn"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Change Style"/>

</LinearLayout>
```

2. Java Code (`MainActivity.java`):

```
package com.example.weatherapp;

import android.graphics.Color;
import android.graphics.Typeface;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    TextView weatherText;
    Button changeBtn;
```

```
boolean changed = false;

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    weatherText = findViewById(R.id.weatherText);
    changeBtn = findViewById(R.id.changeBtn);

    changeBtn.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {

            if (!changed) {
                weatherText.setTextColor(Color.BLUE);
                weatherText.setTypeface(null, Typeface.BOLD_ITALIC);
                weatherText.setText("Weather Updated 🌞 ");

                Toast.makeText(MainActivity.this,
                    "Style Changed!", Toast.LENGTH_SHORT).show();

                changed = true;
            } else {
                weatherText.setTextColor(Color.BLACK);
                weatherText.setTypeface(null, Typeface.NORMAL);
```

```
            weatherText.setText("Weather Information");


            Toast.makeText(MainActivity.this,
              "Back to Default!", Toast.LENGTH_SHORT).show();


            changed = false;
          }
        }
      });
  }
}
```

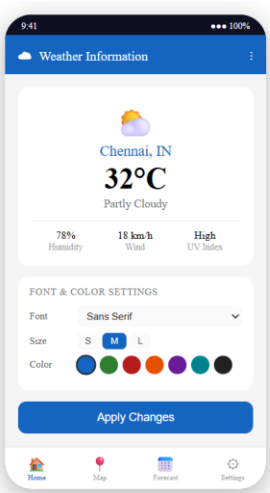
3. AndroidManifest:

```
<application
  android:allowBackup="true"
  android:label="Weather App"
  android:theme="@style/Theme.AppCompat.Light.DarkActionBar">

  <activity android:name=".MainActivity">
    <intent-filter>
      <action android:name="android.intent.action.MAIN" />
      <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
  </activity>

</application>
```

Output:



RESULT:

An mobile application is developed in Android studio to change the font and color of the text and display toast message when the user presses the button.